Quality-control adjustment assistant

Starting point

When a batch falls outside limits, the correction is calculated by reviewing previous cases and checking which corrections worked in similar situations. That approach works, but it depends heavily on individual judgment, consumes time and leaves room for human error when making the decision.

Practical case summary

- An M18 EXTRAMATE BLANCO batch is produced in tank 270.
- The target viscosity range is 120p-150p and the observed result is 180p.
- Today the correction is decided by searching similar past batches and replicating the applied adjustment.
- With the new assistant, results are entered in a logbook and the system recommends the correction based on the tank and five years of historical data.
- The batch is corrected, brought back within limits and accepted.

What the system achieves

- Avoid human calculation errors.
- Create a unified criterion across people and shifts.
- Increase reaction speed at the moment the correction must be chosen.
- Give the team more confidence in operational decision-making.
- Be especially useful when the correction depends on several raw materials rather than a single parameter.

Current status

Jorge Ruano explains that the logbook or “Pepito Grillo”, developed together with Sergio, already works as a web viewer where off-spec production results are entered and a recommended correction is proposed. The tool is still in testing and needs to be validated across different parameters, both individually and in multiple-correction scenarios.

System screenshots

Motor de Ajustes QC Pinturas

Motor hibrido de diagnostico y recomendacion de ajustes para control de calidad en fabricacion de pinturas

Recomendacion | Historicos similares | Trazabilidad | Administracion | Analitica

6.1 Pantalla principal de recomendacion

C. Acciones de mantenimiento

Limpiar | Exportar a Excel | Ex

A. Datos de fabricacion

NombreProducto

B-200 MATE SUPER +

Color

BLANCO

Seccion

VF-270

KGProduccion

10000,00

Especificaciones aplicables a la seleccion

Especificaciones aplicables a la seleccion

Fuente: master:Ordenes Ajustes.xlsx | Nivel: NombreProducto+Color

propiedad	min
densidad	
viscosidad_poises	
viscosidad_seg	
fc	

D. Ejecucion de calculo

Modo rapido (calculo aproximado y mucho mas agil)

Analizar estado

Resultado

Estado general

FUERA_ESPECIFICACION

Incumplimientos

1

propiedad	actual	min	max
densidad	1.52	1.5	1.56
viscosidad_poises	100	120	160
viscosidad_seg	0	None	None
fc	0	None	None

Resultado

Estado general

FUERA_ESPECIFICACION

Incumplimientos

1

propiedad	actual	min	max
densidad	1.52	1.5	1.
viscosidad_poises	100	120	1
viscosidad_seg	0	None	No
fc	0	None	No

Recomendacion de ajuste

Probabilidad exito

100.00%

Fiabilidad

BAJA

MateriaPrima	Cantidad_por_
900131 - AMBROSIA 900131 (SILO E11 LLAVE E-10)	
901009 - PARAGONTA 901009	

Recomendacion basada en ajustes historicos observados en Excel para esta referencia exacta.

Post ajuste estimado: Densidad=1.52, ViscosidadPoises=124.0, ViscosidadSeg=0.0, FC=0.0